Same Storm, Different Boats: Generative AI and the Age Gradient in Hiring[⋆]

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Abstract

Whether generative AI or monetary tightening explains the post-2022 stock-price–job-postings divergence is debated. We study both margins in Sweden using full-population register data (2019–2025) and 4.6 million job advertisements (2020–2026). Using employer-level difference-in-differences and the seven-month gap between the Riksbank's first rate hike (April 2022) and the ChatGPT launch (November 2022), we find that the aggregate posting decline tracks monetary tightening; within employers, employment of workers aged 22–25 in AI-exposed occupations falls 5.5 per cent by mid-2025, concentrated among young women, mainly through reduced hiring and only after ChatGPT.

Keywords: Generative artificial intelligence, Job postings, Labour demand, Employment composition, Monetary policy
JEL: J23, J24, O33

1. Introduction

Since late 2022, US stock prices rose more than 70 per cent while job openings fell more than 30 per cent, prompting debate over whether generative AI or monetary tightening explains the divergence (Thompson, 2025).

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Using ADP payroll data, Brynjolfsson et al. (2025) document a 16 per cent employment decline among young US workers in AI-exposed occupations, the "canaries in the coal mine"; Hosseini Maasoum and Lichtinger (2025) find a similar seniority gradient in US CV and job-posting data. Whether the pattern reflects AI or coincident monetary tightening, and whether it generalises beyond the US, remains open. In Sweden, the broad fall in postings tracks monetary tightening, whereas within employers high-AI-exposure occupations shifted away from workers aged 22–25 after ChatGPT, mainly through lower hiring.

We exploit a timing contrast that helps distinguish the channels: the Riksbank began raising rates in April 2022, seven months before ChatGPT launched in November 2022. If monetary tightening drives the differential posting decline, it should begin with the rate hike; if generative AI drives it, the decline should not appear before late 2022 and may emerge with adoption lags into 2023–2024. Using 4.6 million advertisements from Platsbanken (the Swedish Public Employment Service vacancy portal, 2020–2026), matched to the Dynamic AI Occupational Exposure (DAIOE) index (Engberg et al., 2024)—an occupation-level index of exposure to generative AI—we decompose the aggregate posting decline by AI exposure. Postings across all exposure quartiles peak in mid-2022 and decline together through 2024, with at most a small additional drop in high-exposure occupations after ChatGPT.

We complement the posting evidence with population-register employment data and an employer-level difference-in-differences design (Brynjolfsson et al., 2025). Within the same employers, employment of workers aged 22–25 in high-AI-exposure occupations falls 5.5 per cent (95 per cent confidence interval (CI) [-5.8, -4.9]) below less-exposed occupations by mid-2025. The gradient is concentrated below age 30; the adjustment loads on the hiring margin, not on separations. Beyond the timing contrast, the contribution is to provide population-register evidence of an age-gradient adjustment outside the US, contrasting with the Finnish null in Kauhanen and Rouvinen (2026), which persists even when Swedish workers are reweighted to the Finnish industry-occupation mix.

2. Data and empirical strategy

Job-posting margin. We use the full population of advertisements published on Platsbanken from January 2020 through February 2026. Each ad contains a publication date, a four-digit occupation code from the Swedish Standard

Classification of Occupations (SSYK 2012), a vacancy count, a municipality, and an employer identifier. After removing ads without valid SSYK codes and deduplicating on ad identifiers, 4.6 million ads across 400 occupations remain, aggregated to occupation-by-month cells (26,672 observations).[1]

We measure generative-AI exposure using the DAIOE index (Engberg et al., 2024), which maps AI benchmark capabilities to Swedish occupations at the SSYK 4-digit level. We use the 2023 cross-section of the percentile ranking for generative AI and classify occupations into quartiles, where Q4 is the most exposed.

The baseline specification is

$$\begin{aligned} \ln(\text{postings}_{it}) = \alpha_i + \gamma_t + \beta_1 \cdot \text{PostRB}_t \cdot \text{High}_i \\ + \beta_2 \cdot \text{PostGPT}_t \cdot \text{High}_i + \varepsilon_{it}, \end{aligned} \tag{1}$$

where i indexes SSYK4 occupations, t months, and α_i and γ_t are occupation and month fixed effects. $\text{PostRB}_t = 1$ for $t \geq$ April 2022 (the Riksbank's first rate hike); $\text{PostGPT}_t = 1$ for $t \geq$ December 2022 (the first full month after ChatGPT's 30 November launch); $\text{High}_i = 1$ if occupation i is in the top quartile of generative-AI exposure. Standard errors are clustered at the occupation level. The key test is whether the relative posting decline in high-exposure occupations begins with monetary tightening ($\hat{\beta}_1$) or with the advent of generative AI ($\hat{\beta}_2$).

Employment margin. For employment, we use monthly individual-level employer-declaration data (*Arbetsgivardeklaration på individnivå*) from Statistics Sweden (SCB), hereafter the employer-declaration register. We link each worker to SSYK 4-digit occupation, age, and gender from the annual LISA register (Longitudinal Integration Database for Health Insurance and Labour Market Studies); together they cover the full population of employed individuals.[2] We assign AI exposure from the DAIOE quartile of each worker's most recent occupation. Swedish university completion typically extends into

[1] Historical bulk data (2020–2025) from `https://data.jobtechdev.se/annonser/historiska/`; 2026 data from the real-time JobStream API. Until 2007, private and local-government employers were legally required to post vacancies on Platsbanken (Cronert, 2021); remaining sample-coverage details are in Online Appendix II.4.

[2] Both registers have been stable through the study period: the employer-declaration register since its 2019 introduction, and the SSYK 2012 classification used in LISA throughout.

the late twenties, so workers aged 22–25 proxy very-early-career employment (Online Appendix III.1).

For each age group a we estimate

$$\ln(n_{f,q,t}+1) = \alpha_{f,q}+\delta_{f,t}+\gamma_1\cdot\text{PostRB}_t\cdot\text{High}_q+\gamma_2\cdot\text{PostGPT}_t\cdot\text{High}_q+\varepsilon_{f,q,t}, \quad (2)$$

where f indexes employers, q DAIOE quartiles, and $n_{f,q,t}$ counts age-group-a workers at employer f in quartile q in month t, following Brynjolfsson et al. (2025). Employer×quartile fixed effects $\alpha_{f,q}$ absorb time-invariant compositional differences; employer×month fixed effects $\delta_{f,t}$ absorb all firm-level time-varying shocks, so identification comes from within-employer recomposition across AI-exposure quartiles. Standard errors are clustered by employer×quartile. The coefficient γ_2 is the age-group-specific analogue of β_2 in Equation (1).

3. Results

Figure 1 shows the Swedish stock-price–job-postings divergence. The OMXS30 (the Stockholm large-cap index) reached 175 per cent of its February 2020 level by early 2026, while job postings across all AI-exposure quartiles declined together from the Riksbank rate hike in April 2022. The divergence opens seven months before ChatGPT, coinciding with monetary tightening. The pattern is identical with the broader OMXSPI All-Share index (Online Appendix II.2).

Estimating Equation (1), we estimate the rate-hike interaction precisely at $\hat{\beta}_1 = -0.127$ (95 per cent CI $[-0.203, -0.051]$; $p < 0.01$), while the post-ChatGPT interaction is small and statistically indistinguishable from zero, $\hat{\beta}_2 = -0.062$ (95 per cent CI $[-0.137, +0.013]$; $p = 0.11$). Any additional posting effect after ChatGPT is small and sensitive to trend assumptions (Online Appendix II.4, II.9). AI exposure is uncorrelated with revealed monetary-policy sensitivity ($r = 0.04$, $p = 0.51$; Online Appendix II.7), consistent with the channels being distinct.

The employment margin shows a different pattern. Figure 2 plots the raw monthly employment series by age group and AI-exposure quartile (3-month moving average; the unsmoothed series is in Online Appendix III.1). Employment of workers aged 22–25 in high-AI-exposure occupations diverges from the comparison groups during 2024 and the gap widens through mid-2025; the aggregated 26+ series is comparatively stable, masking heterogeneity

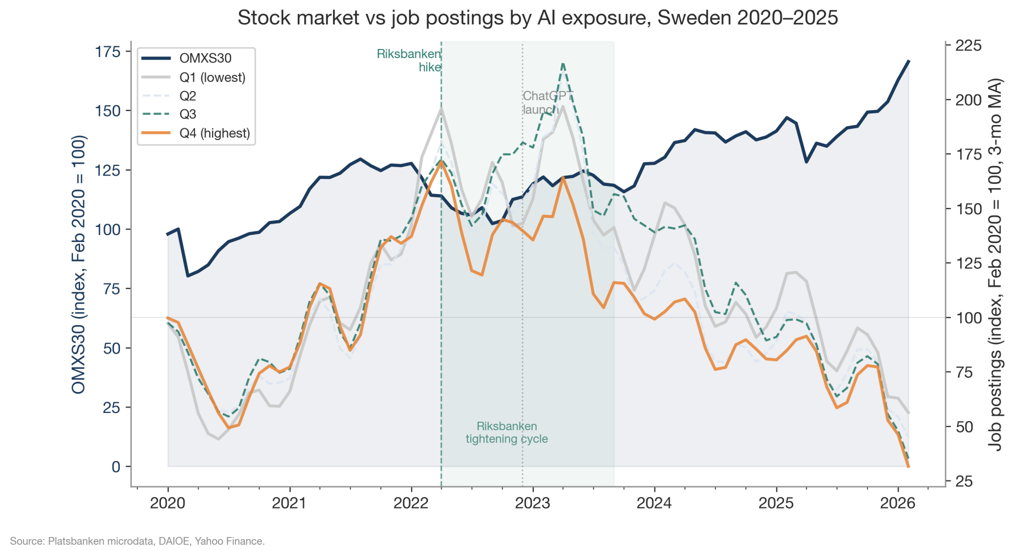

Figure 1: Stock market vs job postings by AI-exposure quartile, Sweden 2020–2026. Daily OMXS30 closing prices (Yahoo Finance) are resampled to monthly averages and indexed to 100 at February 2020 (left axis); Platsbanken posting index by generative-AI exposure quartile is shown on the right axis. Vertical lines mark the Riksbank's first rate hike (April 2022) and the ChatGPT launch (November 2022).

within older bins.

Figure 3 shows the half-year event study. Our headline estimand is the 22–25 event-study endpoint at mid-2025; the average DiD and Poisson PML coefficients reported below are supporting checks. By the first half of 2025, employment of 22–25 year olds in top-quartile AI-exposed occupations is 5.5 per cent below less-exposed occupations within the same employers (95 per cent CI [-5.8, -4.9]). Averaged across all post-ChatGPT periods (a supporting check that pools the flat 2023 with the late-period decline), the difference-in-differences coefficient is $\hat{\gamma}_2 = -0.010$ (95 per cent CI [-0.014, -0.006]; $p < 0.001$). Pre-trend tests reject strict parallel trends, but the estimated pre-period deviations for the 22–25 and 26–30 groups are positive, biasing the post-ChatGPT estimate *toward zero*: the -5.5 per cent endpoint is, if anything, conservative; the modest 50+ rise should be read with the opposite sign of caution. Poisson pseudo-maximum likelihood (PML) yields a multiplicative effect of -16 per cent for 22–25 (95 per cent CI [-18, -14]) across four specifications. Both estimators agree on the headline 22–25 finding;

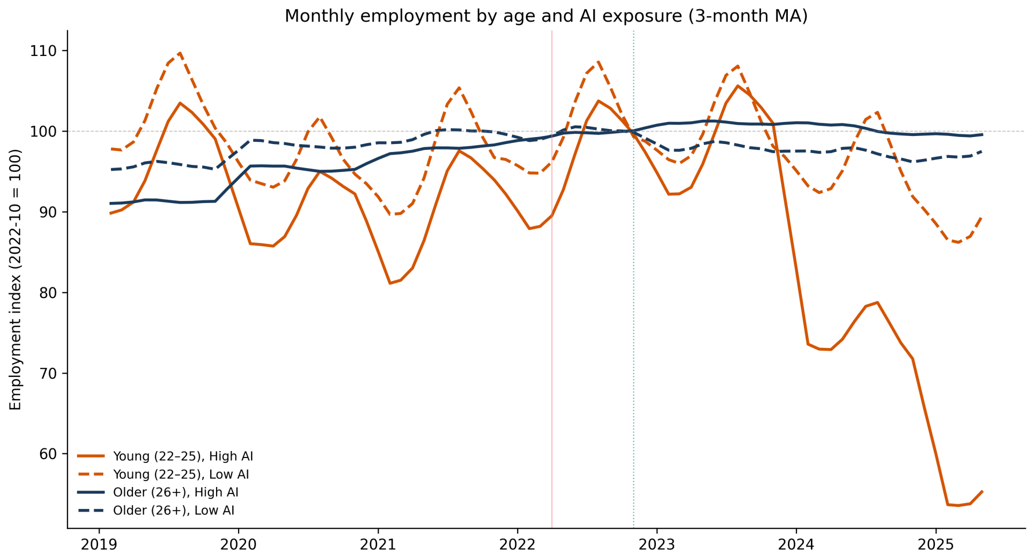

Figure 2: Monthly employment by age group and AI exposure, Sweden 2019–2025 (3-month moving average). Employer-declaration register data indexed to October 2022 = 100. Young (22–25) versus older (26+) workers, split by AI-exposure quartile (DAIOE). Vertical lines mark the Riksbank's first rate hike (April 2022) and the ChatGPT launch (November 2022).

the magnitude gap reflects a scale difference between $\ln(y+1)$ and Poisson (Online Appendix III.6).

The decline operates almost entirely through the hiring margin. The hires coefficient for ages 22–25 is $\hat{\gamma}_2^H = -0.0051$ (95 per cent CI $[-0.0061, -0.0041]$; $p < 0.001$), while the separations coefficient is one quarter as large in absolute terms ($\hat{\gamma}_2^S = -0.0012$, 95 per cent CI $[-0.0020, -0.0004]$; $p < 0.01$). Under Sweden's last-in-first-out employment-protection legislation (LAS), separations of incumbent workers are constrained, so adjustment loads on the hiring margin: high-exposure employers reduce entry hiring rather than dismissing incumbents (Online Appendix III.3). This pattern is consistent with generative AI compressing codified, routine entry-level tasks rather than displacing established workers.

The age profile is sharply concentrated below age 30. The gradient extends to ages 26–30 with a lag: the half-year event-study endpoint reaches -0.049 by mid-2025, although the average DiD coefficient pools this late effect with a flat 2022–2023 and is null on average (Online Appendix III.2). Effects for ages 31–49 are essentially zero. The 50+ point estimate is positive but small ($\hat{\gamma}_2 = +0.010$, 95 per cent CI [+0.008, +0.012]) and is sensitive both to pre-trend slopes and to the choice of exposure measure: it disappears entirely under the Eloundou et al. (2024) score (Online Appendix II.5, III.7). Population-level employment remains broadly stable, consistent with occupational reallocation rather than net job destruction. The 22–25 result strengthens in the long-tenured subsample, ruling out staleness in occupational coding as a driver (Online Appendix III.8).

The decline is concentrated among young women: $\hat{\gamma}_2 = -0.016$ (95 per cent CI [−0.020, −0.012]) for women aged 22–25, versus −0.007 (95 per cent CI [−0.013, −0.001]) for men, with a triple-difference $p < 0.01$ (Online Appendix III.5). Reweighting women's occupational distribution to match men's narrows the gap by about two-fifths, leaving a residual within-occupation component.

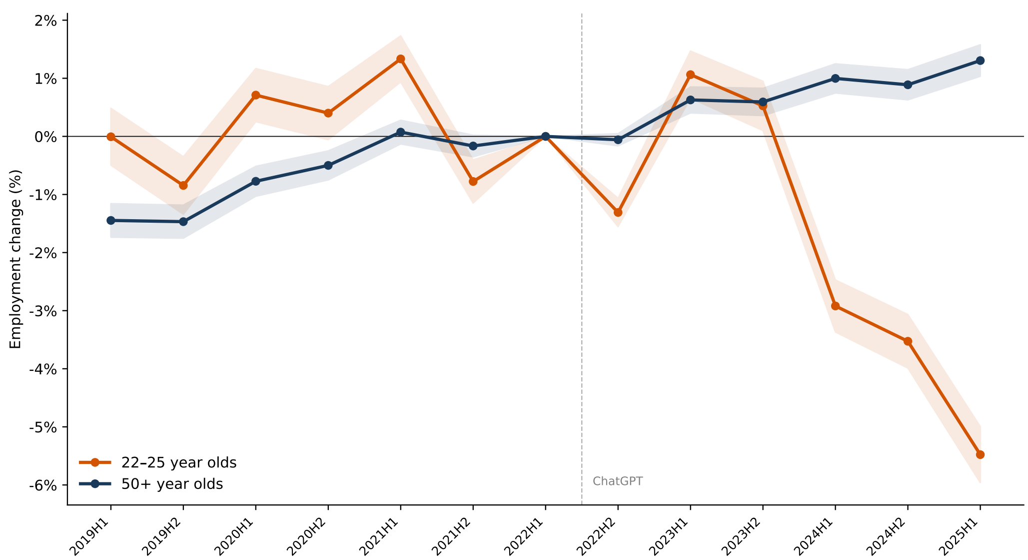

Figure 3: Event study of employment effects by age group and AI exposure, ages 22–25 and 50+. Coefficients from employer-level regressions with employer×quartile and employer×month fixed effects, half-year periods, reference: first half of 2022. Shaded areas show 95 per cent confidence intervals. The dashed line marks the ChatGPT launch (November 2022). Coefficients from the log specification approximate percentage changes. All six age groups appear in Online Appendix III.2.

4. Conclusion

The age gradient is consistent with generative AI substituting for codified, routine entry-level tasks while complementing the tacit, experiential expertise of senior workers (Acemoglu and Restrepo, 2019; Acemoglu et al., 2026). Adjustment loads on the hiring margin, consistent with Sweden's incumbent-protection legislation. We do not directly observe employers' AI adoption, so unobserved organisational changes correlated with exposure cannot be ruled out. If the pattern persists, the displacement of entry-level cohorts raises a longer-term skill-formation concern; transition pathways and junior-role redesign deserve attention.

Declaration of competing interest

The authors declare no competing interests.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the authors used Claude (Anthropic) to assist with code development, data processing, and editorial suggestions. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

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Data availability

Platsbanken job advertisements are publicly available under a CC0 licence (`https://data.jobtechdev.se/annonser/historiska/`; `https://jobstream.api.jobtechdev.se`). The employment analysis uses confidential individual-level administrative microdata from Statistics Sweden (SCB), accessed via the secure MONA platform under ethical approvals listed below; access procedures are documented in Online Appendix IV.1. All replication code is publicly available at `https://github.com/Magnus-L/canaries-sweden`; an archived release is referenced in the Online Appendix.

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